Final integrative project

End-to-end time-series pipeline

Statistics · Decomposition · ARMA · GARCH · VaR

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Context and objectives

This integrative project mobilises the notions of every chapter of the module:

- **Ch. 1 Basic statistics**: descriptive analysis, confidence intervals, normality tests.
- **Ch. 2 Decomposition**: classical time-series decomposition.
- **Ch. 3 Stationarity**: ACF/PACF and Ljung–Box.
- **Ch. 4 Non-stationarity**: ADF/KPSS unit-root tests and differencing.
- **Ch. 5 ARMA**: Box–Jenkins identification, MLE, forecasting.
- **Ch. 6 ARCH/GARCH**: Engle LM, GARCH$(1,1)$, and Value-at-Risk.

The fil-rouge task is a complete *risk-management study* of a synthetic but realistic daily stock-price series. The goal is to produce, by the end of the project, a one-day Value-at-Risk that is well-calibrated according to a Kupiec back-test.

Setup

- Language: Python 3, dependencies: `numpy`, `pandas`, `matplotlib`, `scipy`, `statsmodels`, `arch`.
- Reproducibility: `numpy.random.seed(42)`.
- Skeleton in `seed/projet.py`; full reference solution in `correction/projet.py`.
- Dataset: `data/stock_prices.csv` – $T = 2000$ synthetic daily prices with drift and GARCH volatility, regenerated by `generate_dataset.py`.
- Deliverables:
  - a reproducible `projet_<name>.py`;
  - a 10–12 page report PDF with figures and discussion;
  - a 12-min individual oral defence.

Data

The data file `data/stock_prices.csv` contains $T = 2000$ daily closing prices P_t over the period 2018-01-02 to 2025-09-15 (business days). The series is generated as

$$\log P_t = \log P_{t-1} + \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1),$$

with $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ – a textbook GARCH$(1,1)$ data-generating process on the log-returns. The true parameters are kept hidden; you will recover them from the data.

Exercise 1 – Part A – Basic statistics on returns (Ch. 1)

(A.1) Load `data/stock_prices.csv`. Compute the log-returns $r_t = \log(P_t/P_{t-1})$, $t = 1, \dots, T-1$.
(A.2) Report the sample mean $\widehat{\mu}$, sample standard deviation $\widehat{\sigma}$, sample skewness, sample excess kurtosis. Provide a 95% confidence interval for the mean (using the Student-t approximation).
(A.3) Test the null hypothesis $H_0 : \mu = 0$ at the 5% level. Conclude.
(A.4) Plot a histogram of r_t together with the $\mathcal{N}(\widehat{\mu}, \widehat{\sigma}^2)$ density. Run a Jarque–Bera normality test and comment.

Exercise 2 – Part B – Decomposition of the price level (Ch. 2)

(B.1) On the price series P_t (*not* the returns), extract the trend by a centred moving average of width $k = 50$ business days.
(B.2) On the log-price $\log P_t$, fit a linear trend by OLS:

$$\log P_t = \beta_0 + \beta_1 t + u_t.$$

Report $\widehat{\beta}_1$ – the implied mean log-return. Compare with the sample mean of r_t (Part A).
(B.3) Plot P_t, the moving-average trend and the OLS exponential trend on the same figure. Comment in 5 lines.

Exercise 3 – Part C – Stationarity diagnostics (Ch. 3)

(C.1) Plot the sample ACF of $\log P_t$ up to lag 60. What do you observe? Is the series consistent with a stationary process?
(C.2) Same on the returns r_t. Are returns serially uncorrelated?
(C.3) Run the Ljung–Box test on r_t at lags $m \in \{10, 20\}$. Conclude.
(C.4) Run the Ljung–Box test on r_t^2 at the same lags. What do the two results, taken together, suggest?

Exercise 4 – Part D – Unit-root testing (Ch. 4)

(D.1) Run the ADF test on $\log P_t$ with `regression="ct"` (constant + trend) and on r_t with `regression="c"`. Report stat, p-value, decision at 5% in a small table.
(D.2) Run the KPSS test on the same two series.
(D.3) Build the ADF/KPSS combination matrix on $\log P_t$. In which cell does it land? What does this imply about the integration order $I(d)$ of the price series?
(D.4) Confirm: $\log P_t \sim I(1)$ and $r_t \sim I(0)$.

Exercise 5 – Part E – ARMA modelling (Ch. 5)

(E.1) On the returns r_t, plot the sample ACF and PACF up to lag 20. Identify any structure in the conditional mean.
(E.2) Fit three candidates by MLE (`statsmodels.tsa.arima.ARIMA`):

- ARMA$(0,0)$: constant mean only;
- ARMA$(1,0)$;
- ARMA$(0,1)$.

Report a comparison table: log-likelihood, AIC, BIC, Ljung–Box(24) p-value.
(E.3) Choose the best model on the (AIC, BIC, Ljung–Box) triplet. The constant-mean model often wins – justify why this is expected for daily returns.

Exercise 6 – Part F – GARCH$(1,1)$ and VaR (Ch. 6)

(F.1) On the residuals $\widehat{\varepsilon}_t$ of the chosen mean model from Part E, run Engle's LM test at lags $5, 10, 20$. Confirm the presence of ARCH effects.
(F.2) Fit a GARCH$(1,1)$ on the returns r_t with a constant mean and Gaussian innovations using the `arch` package. Report $\widehat{\mu}$, $\widehat{\alpha}_0$, $\widehat{\alpha}_1$, $\widehat{\beta}_1$ with their standard errors. Compute $\widehat{\alpha}_1 + \widehat{\beta}_1$ and the implied unconditional variance.
(F.3) Compute the standardised residuals $\widehat{z}_t = \widehat{\varepsilon}_t / \widehat{\sigma}_t$. Run Engle's LM test on $\widehat{z}_t^2$. Has the GARCH absorbed the volatility clustering?
(F.4) Compute the conditional one-day Value-at-Risk

$$\mathrm{VaR}_{T+1}(\alpha) = -(\widehat{\mu} + z_\alpha \, \widehat{\sigma}_{T+1}), \qquad \alpha \in \{1\%, 5\%\}.$$

Compare with the unconditional VaR (using the sample $\widehat{\sigma}$).

Exercise 7 – Part G – Kupiec back-test (synthesis)

(G.1) Split the data: train = first 1750 obs, test = last 250 obs.
(G.2) On the test set, compute a *rolling* conditional VaR at $\alpha = 5\%$: at each day t in the test set, fit the GARCH on observations $1..t-1$, compute $\widehat{\sigma}_t$, and predict $\mathrm{VaR}_t(5\%)$.
(G.3) Count the number of *exceedances* – days where the realised loss $-r_t$ exceeded $\mathrm{VaR}_t(5\%)$.
(G.4) Run the Kupiec proportion-of-failures test:

$$LR_{\text{Kupiec}} = -2 \log \left[\frac{(1-\alpha)^{N-X} \alpha^X}{(1-\widehat{\pi})^{N-X} \widehat{\pi}^X} \right] \xrightarrow{d} \chi^2_1,$$

with $N = 250$, X the number of exceedances, $\widehat{\pi} = X/N$. Reject the null of correct unconditional coverage if $LR > \chi^2_{1,0.95} = 3.841$.
(G.5) Conclude: is the GARCH-VaR well-calibrated?

Marking grid (out of 100)

Criterion	Points
Part A: descriptive statistics + Jarque–Bera	10
Part B: decomposition + linear-trend fit	10
Part C: ACF/PACF + Ljung–Box on r_t and r_t^2	10
Part D: ADF/KPSS + combination matrix	15
Part E: ARMA candidates + AIC/BIC/LB	15
Part F: GARCH$(1,1)$ + standardised diagnostics + VaR	20
Part G: Kupiec back-test	10
Reproducibility + report quality	5
Oral defence	5
Total	**100**

Bonus. Student-t innovation distribution in the GARCH (+5 points), Christoffersen's conditional-coverage test on top of Kupiec (+5), comparison with EGARCH or GJR-GARCH (+5).

Submission

- **Code**: `projet_<name>.py` (or `projet_<name>.ipynb`) reproducible with seed 42.
- **Report**: 10–12 PDF pages including:
  - summary of the dataset and stylised facts;
  - all required tables and figures (one figure per part);
  - Kupiec back-test result;
  - discussion of model choice and calibration.
- **Oral defence**: 12 minutes per student, 5 minutes of questions.

Warning

Install once: `pip install numpy scipy matplotlib pandas statsmodels arch`. The full reference pipeline runs in ≈ 15 seconds on a laptop CPU; the rolling Kupiec back-test (Part G) is the slowest part (~ 30 s).

Good luck!

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